Name	Shiven Saxena
Student ID	34997407
Studio Session	17

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.
Tip: Test your URL in incognito mode to ensure it is a working URL

https://ssax0009-ctrl.github.io/DV2_FIT3179/

Part B: Domain Description

Describe your domain chosen for DV II

- The domain for this visualisation is: Australian women's healthcare, and how hospital care for hysterectomy and endometriosis is affected by state, local hospital network, funding sector and age.
- The users are health-service planners and women's health advocates, who understand complex terms such as median wait, LHN, SA3, age-standardised rate, SES group and suppressed data.
- Their workflow is constrained by public-only waiting times, waits and rates from different years, and withheld values that must not be read as short waits.

Describe the Why and Who aspects of your DV II

Why
The goal is to survey how affordable and accessible Women's health is through looking at the statistics of the two of the most common reasons for hospitalisations Endometriosis and Hysterectomy.

Who (in the perspective of Australians):

- Health Service Planners and Policy makers:
    - We analyse to the local level and show how this differs from the state level, integral for local governments to know how to "plan ahead"
- Women Health advocates:
    - Showing trends and how accessibility is improving over time, especially when different groups are in perspective
    -
- General Women (especially those with Endometriosis or Hysterectomy):
    - Analyse the how accessibility for treatment and where is ready for them

Describe the "What" aspect of your DV II

Australian atlas of healthcare variation time series 2014–15 to 2021–22: Hysterectomy hospitalisations

Data name	**Author**	**Use**	**Why?**
Hysterectomy hospitalisations, 2014–15 to 2021–22	AIHW Australian Atlas of Healthcare Variation (National Hospital Morbidity Database, ABS population estimates). Women 15+, age-standardised to the 2001 population.	State and SA3 rates, Indigenous status, remoteness and SES, private share (Figures 3–11)	Hysterectomy is a very common surgery and can be described as a benchmark for all surgeries, this data is comprehensive on locations, as well as sectors.
Elective surgery waiting times, 2024–25	AIHW *Australian hospital statistics*, Table 4.7 for state medians, plus LHN files with coordinates	Figures 1–2	Observing wait times shows us how accessible a procedure is
Endometriosis hospital procedures (PHE 357)	AIHW, 2026 supplementary tables	Admissions, urgency, age, sector (Figures 12–15)	Most of the data for endometriosis which is one of the most prevalent illness specific to women

Boundaries	ABS ASGS Edition 3 (SA3); geoBoundaries gbOpen (state outlines)	Maps	For making maps

Some notes on the Creation Process:

- Missing Data was often an issue, especially when 2 graphs are complementary, so I made the difficult decision to suppress both in order to prevent misleading information
- Some data was not all in the same place, such as coordinates, so I had to find that separately.
- One of the biggest problems was quality data, as there were not many actual data providers. Weighing the odds, I decided to use old data 2021 - 2022 that is credible over non credible data that is relevant.

Describe the "How" aspect of your DV II

Figure	Idiom	Reasoning
1	Choropleth map	Help show density of state, and can be used to compare for outliers (figure 2)
2	Proportional symbol map	Shows the movement of densities (closer to the coast the more hospitals for example) help compare with figure 1
3	Ridgeline graph	Shows the distribution of women for hospitalisations, allows us to see how location affects the average for each state (exposing lying with state averages)
4	Interval plot	Similar to ridgeline, but further shows exemplifies the idea of the outliers.
5	Dot Map	Shows that variation is spread geographically and what regions are most sparse
6	Radial chart	Combines two factors at once: remoteness by colour and SES by slice. It demonstrates that the highest rates are not in the most remote areas.

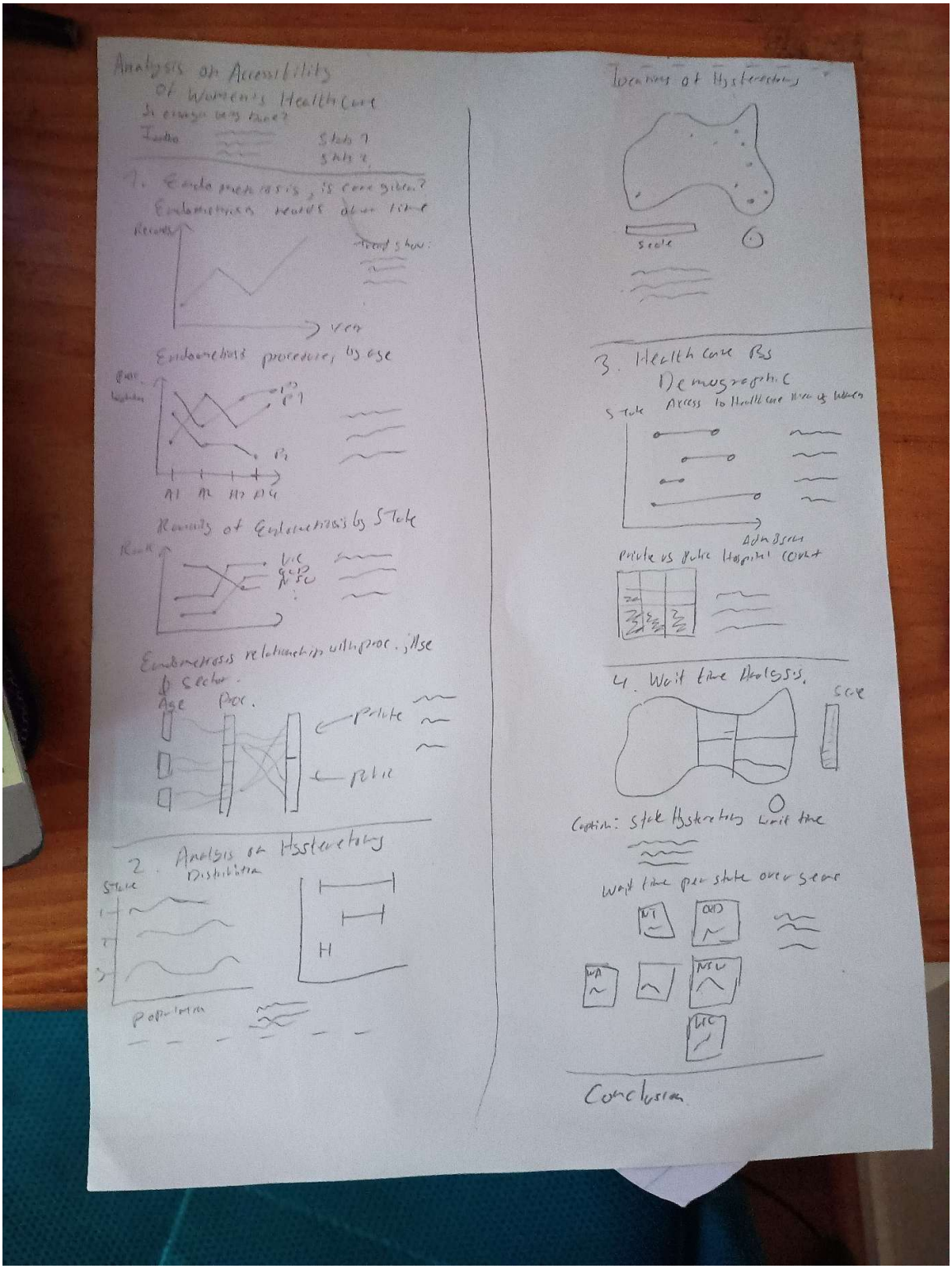
7	Dumbbell	Two groups (First nation and Others) per state with the gap drawn, so the direction of the difference is the first thing seen. Helps us compare if First Nation people are getting the care they deserve
8	Waffle	One square equals 1%, so private share is counted, not estimated. An easy visual to see how private hospitals dominate over public hospitals.
9	Slope chart	Compares 2 main years, allows us to see whether the surgery admissions increase or decrease.
10	Tile-grid map with sparklines	Equal-size tiles stop Tasmania, the ACT and the NT from vanishing on a real map. Sparklines add the shape between the endpoints. A fun intuitive map that shows rates per state.
11	Heatmap	Another usage of distribution, but here we compare state and time. Allows us to see a projection of how hysterectomy records are falling
12	Time series	Shows growth/decline of endometriosis rate.
13	waterfall	Demonstrates ideas such as medical urgency as well as assisting the time series
14	Parallel Plot	Ten procedures are muted for context. Two (laparoscopic excision and cystoscopy) are emphasised to show different age peaks and how age plays a role on surgery given and accessibility.

Special Features:
Used a button for glossary for ease of use.

AI Declaration (also on visualisation):
AI was used in this assignment with LLMs such as Claude, Copilot, Chatgpt and Gemini to help debug code, ease of code format and structuring as well as giving advice.

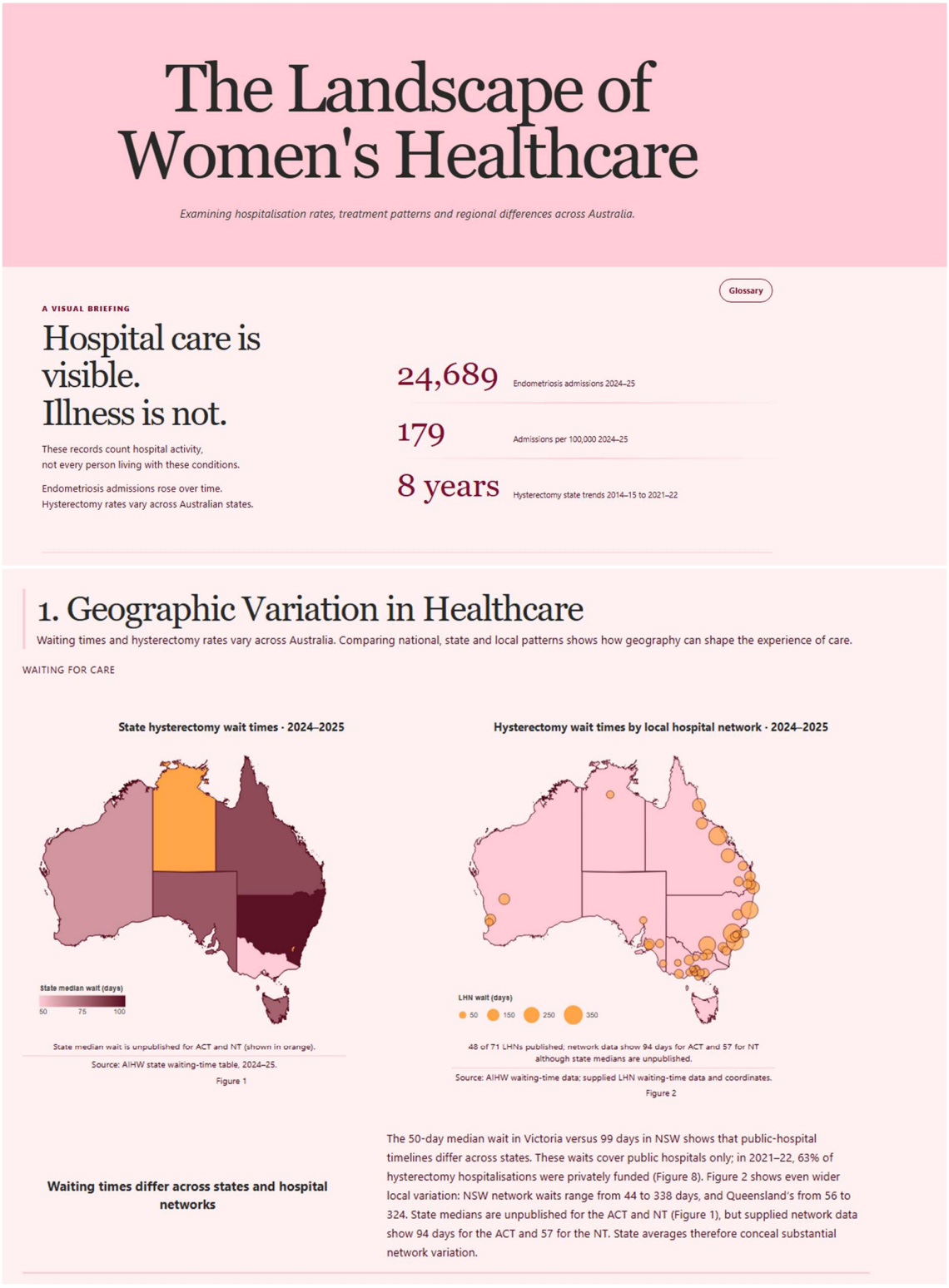
Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.
https://github.com/ssax0009-ctrl/DV2_FIT3179/blob/main/sketch.pdf

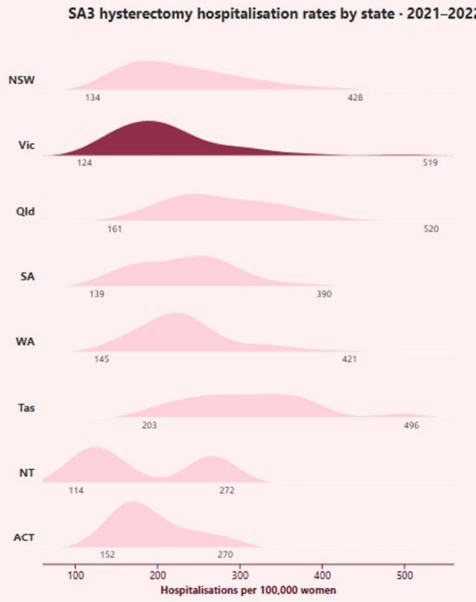

Analysis on Accessibility
of Women's Health Care
Locations of Hysterectomy
Scale
Endometriosis procedures by ese
3. Health Care Bs
Demographic
Access to Health Care
Private vs Public Hospital Count
4. Wait time Analysis
Scale
Caption: State Hysterectomy Wait time
Wait time per state over years
2. Analysis on Hysterectomy
Distribution
State
Population
Conclusion

Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here



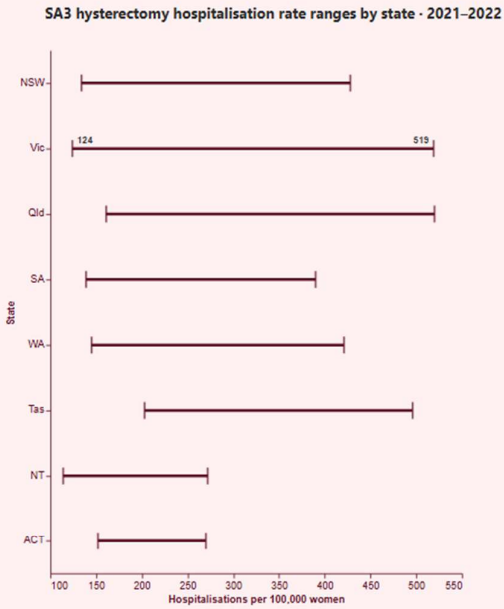
VARIATION WITHIN STATES



Each ridge smooths the SA3 rates in one state; numbers mark its lowest and highest area. Suppressed rates are omitted, and the ACT and NT curves rest on only 5 to 6 areas.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 3



24 SA3 values are suppressed; statewide counts of valid SA3s range from 5 to 87.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 4

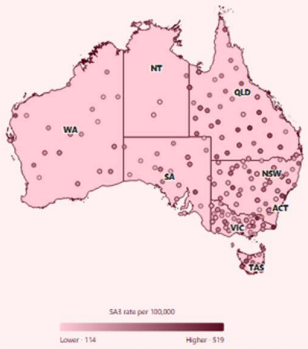
State averages conceal substantial local variation

Figures 3 and 4 show that areas within the same state can have very different hysterectomy hospitalisation rates. In Victoria, the range runs from 124 per 100,000 women in Whitehorse–West to 519 in Ballarat. The key story is not simply that states differ, but that the place someone lives within a state may also matter.

Local differences become visible on the map

Figure 5 adds a geographic dimension to the ranges shown in Figure 4: it shows where the measured SA3 areas sit, rather than only how high or low their rates are. The spread across the map reinforces that variation is not confined to one state or one region. The dots are schematic, however, so their positions show distribution rather than exact locations.

Locations of SA3 areas with published hysterectomy hospitalisation rates · 2021–2022



Selected areas · rate per 100,000 · hospitalisations · female population

Ballarat: 519 · 235 hospitalisations · population 48,988
Whitehorse - West: 124 · 56 hospitalisations · population 47,554
Darwin City: 114 · 12 hospitalisations

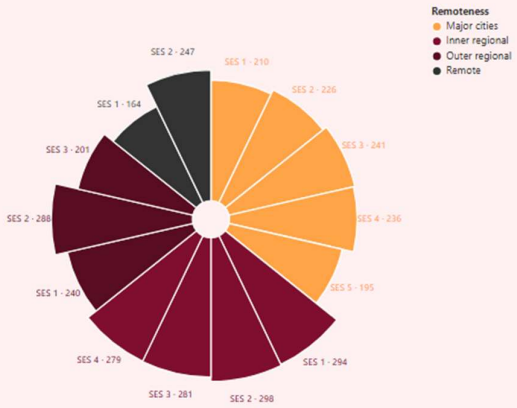
316 rates shown; 24 are suppressed and state totals range from 5 to 87. Locations are schematic.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 5

REMOTENESS AND SOCIOECONOMIC CONTEXT

Hysterectomy hospitalisation rates by remoteness and SES group · 2021–2022



The highest rates are not found in the most remote areas

In Figure 6, inner regional areas record 292 hospitalisations per 100,000 women, above major cities at 218 and remote areas at 193. This challenges a simple assumption that remoteness alone explains differences in care. The socioeconomic slices also show variation within remoteness categories, suggesting that geography and social circumstances should be considered together rather than as competing explanations.

Area: age-standardised rate/100k; colour: remoteness; slices: SES; Outer Regional/Remote SES suppressed.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 6

2. Healthcare Patterns by Demographics

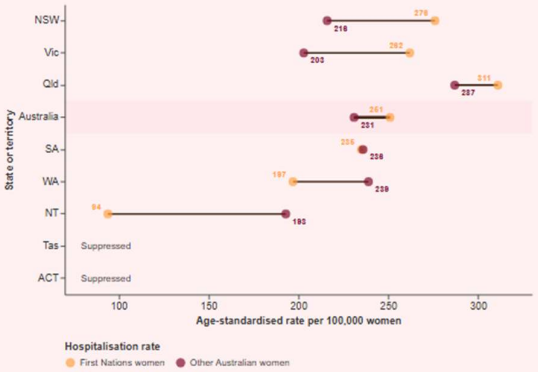
Differences in rates, waiting times and funding show that healthcare experiences are not uniform. Comparing these measures helps identify where further questions about access and service delivery arise.

FIRST NATIONS WOMEN AND HYSTERECTOMY RATES

National averages hide different state patterns

In 2021–22, First Nations women had a higher national hysterectomy hospitalisation rate than Other Australian women (251 versus 231 per 100,000). However, Figure 7 shows contrasting state-level patterns: First Nations women had higher rates in NSW, Victoria and Queensland, while Other Australian women had higher rates in WA and the NT. These results highlight how national averages can conceal geographic variation in recorded hysterectomy hospitalisations.

Hysterectomy hospitalisation rates by Indigenous status and state · 2021–2022



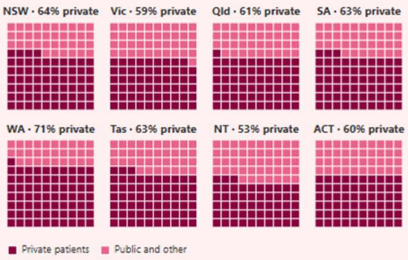
Hospitalisation rates are not measures of prevalence or need; Tasmania and ACT are suppressed.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 7

PUBLIC AND PRIVATE FUNDING

Private-patient share of hysterectomy hospitalisations by state · 2021–2022



Each square is 1% of a state's hysterectomy hospitalisations; nationally, 63% were private.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 8

Private hospitals account for most hysterectomy admissions

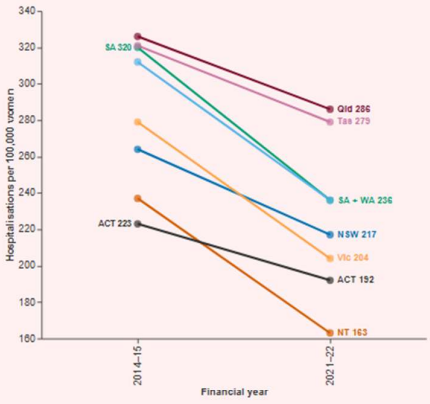
Figure 8 shows that 63% of hysterectomy hospitalisations were privately funded nationally. The share ranges from 53% in the NT to 71% in WA, making the private sector the larger part of recorded activity in every state shown. This uneven funding mix matters because national totals can make the system look more uniform than the state-level picture suggests.

3. Temporal Trends in Hysterectomy Rates

Hysterectomy rates generally declined over the period, but the pace and shape of change differed by state. Looking at the full trajectory reveals more than a single national trend.

CHANGE OVER TIME

Hysterectomy hospitalisation rates by state · 2014–2022

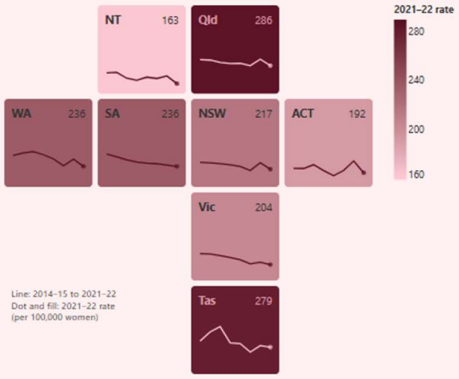


Age-standardised rates; SA and WA finish at the same rate.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2014–15 and 2021–22.

Figure 9

Hysterectomy hospitalisation rates by state · 2014–2022



Fill shows each state's 2021–22 age-standardised rate per 100,000 women; each line runs from 2014–15 to 2021–22.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2014–15 to 2021–22.

Figure 10

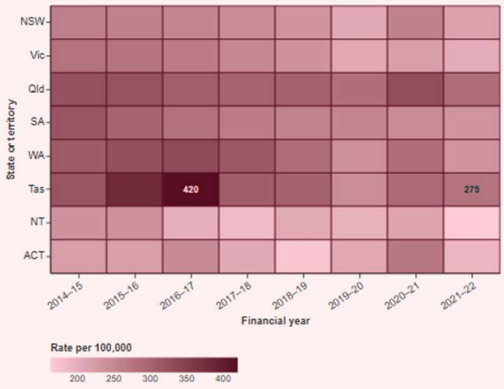
Declining rates did not erase differences between states

Figure 9 shows broad declines in hysterectomy hospitalisation rates, while Figure 10 shows how trajectories differed between states. Comparing only the 2014–15 and 2021–22 endpoints, the gap between the highest and lowest rates widened from 103 to 123 per 100,000 women. The gap was wider in between: in 2016–17, Tasmania's rate of 420 versus the NT's 200 created a 220-point difference (Figure 11). The endpoints alone therefore miss important variation over time.

Tasmania followed a different trajectory

Figure 11 highlights Tasmania's rise to 420 hospitalisations per 100,000 women in 2016–17, followed by a fall to 279 in 2021–22. That peak stands out against the broader pattern of declining rates and shows why a start-to-end comparison can miss important changes in between. The trajectory is unusual, but the data alone do not identify its cause.

Annual age-standardised hysterectomy hospitalisation rates by state · 2014–2022



Changes around 2019–20 to 2021–22 overlap with COVID-era restrictions.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2014–15 to 2021–22.

Figure 11

4. Endometriosis Hospitalisations and Procedures

Endometriosis hospitalisations have increased, while procedure patterns differ by age and hospital sector. Together, these charts show how recorded treatment is distributed and not the full prevalence of the condition.

GROWTH IN HOSPITAL ACTIVITY

Annual endometriosis admissions and crude admission rates · 2015–2025

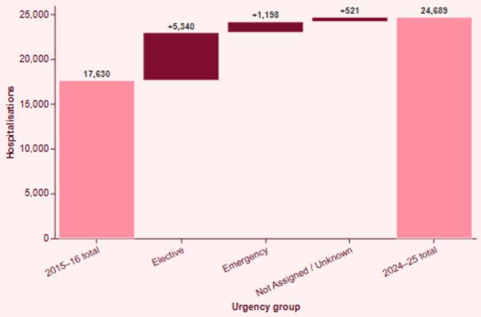


Crude rates are not age-adjusted and are not a measure of disease prevalence.

Source: AIHW PHE 357, Endometriosis hospital procedures, 2015–16 to 2024–25.

Figure 12

Change in endometriosis admissions by urgency · 2015–2025



The waterfall compares total admissions at the start and end of the series.

Source: AIHW PHE 357, Endometriosis hospital procedures, 2015–16 and 2024–25.

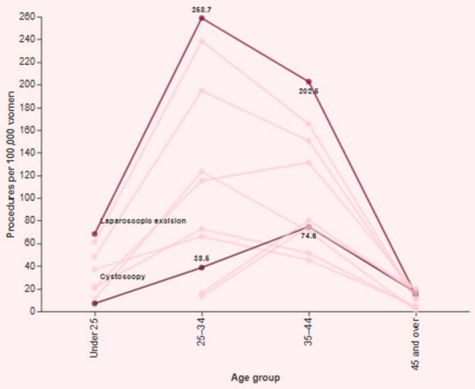
Figure 13

The rise in endometriosis admissions was driven mainly by elective care

Figure 12 shows admissions rising from 17,630 to 24,689 between 2015–16 and 2024–25, an increase of about 40%. Figure 13 indicates that elective admissions account for around 76% of that growth. The increase is therefore concentrated in planned hospital care, making elective treatment an important part of the story behind rising recorded activity.

AGE AND PROCEDURE TYPE

Selected endometriosis procedure rates by age group - 2024–2025



Rates are for selected procedures in 2024–25, not unique people.

Source: AIHW PHE 357, Endometriosis hospital procedures, 2024–25.

Figure 14

Procedure patterns peak during the main reproductive years

In Figure 14, six of ten selected procedures peak at ages 25–34 and four at ages 35–44. This pattern is consistent with endometriosis care being concentrated during reproductive years. Symptoms often improve after menopause, although the condition can persist. The decline in older age groups should therefore be read as a pattern in recorded procedures, not evidence that endometriosis disappears after a particular age.

5. Key Findings and Implications

The data reveal variation in care, but not its full explanation

Across Figures 1–15, place is associated with waiting times and hysterectomy rates measured in different years; state-level patterns also vary by Indigenous status and funding sector. Meanwhile, endometriosis hospitalisations have grown, particularly through elective care, and recorded procedures are concentrated in private hospitals and reproductive-age groups. Together, these findings show that healthcare activity is unevenly distributed. Explaining whether those differences reflect need, access or service organisation requires evidence beyond hospital records alone.

TERMS TO KNOW

A short glossary.

AIHW
Australian Institute of Health and Welfare, the national agency that publishes health and welfare statistics.

Age-standardised rate
A rate adjusted to account for differences in age between populations, making comparisons fairer.

Crude rate
The number of events relative to a population, without adjusting for its age profile.

Endometriosis
A condition where tissue similar to the lining of the uterus grows elsewhere in the body.

Hysterectomy
An operation to remove the uterus.

LHN
Local Hospital Network: a group of public hospitals and health services serving a region.

Median wait time
The middle wait time: half of the patients waited less, and half waited longer.

SA3
Statistical Area Level 3, a geographic area used by the Australian Bureau of Statistics to report regional data.

SES
Socioeconomic status: a way of describing social and economic circumstances, such as relative advantage or disadvantage.

Suppressed data
Figures not published by the data source, often to protect privacy or because estimates are unreliable.

SOURCES & METHODS

Built from published Australian data.

Australian Institute of Health and Welfare. (2026). *Endometriosis hospital procedures* (Cat. no. PHE 357) [Supplementary data tables].

Australian Institute of Health and Welfare. (n.d.). *Australian atlas of healthcare variation time series 2014–15 to 2021–22: Hysterectomy hospitalisations* [Data set]. Project-supplied workbook. Source notes identify the National Hospital Morbidity Database and Australian Bureau of Statistics estimated resident populations.

Australian Institute of Health and Welfare. (n.d.). *Elective surgery waiting times 2024–25: Australian hospital statistics* [Data set]. Project-supplied workbook. Table 4.7 was used for state-level procedure waits. Accompanying project-supplied hospital and Local Hospital Network (LHN) files contain the mapped wait values; LHN records with coordinates were supplied for mapping, and values marked NP or left blank are shown as unavailable.

Australian Bureau of Statistics. (2021). *Australian Statistical Geography Standard (ASGS): Edition 3, July 2021–June 2026*. https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs/edition-3-july-2021-june-2026

Runfola, D., Anderson, A., Baier, H., Crittenden, M., Dowker, E., Fuhrig, S., Goodman, S., Grimsley, G., Layko, R., Melville, G., Mulder, M., Panganiban, J., Peck, A., Seitz, L., Shea, S., Slevin, H., Youngerman, R., & Hobbs, L. (2020). geoBoundaries: A global database of political administrative boundaries. *PLOS ONE, 15*(4), e0231866. https://doi.org/10.1371/journal.pone.0231866 (gbOpen Australia ADM1 boundary data).

Hysterectomy rates cover women aged 15 and over and are age-standardised to the 2001 Australian population. Waiting-time measures are days waited by admitted patients; state percentiles are unpublished for the ACT and NT. Endometriosis rates are crude rates reported by AIHW. The ACT did not provide private-hospital data for 2024–25.

Visualisation by Shiven Saxena · 2026 · AI assistance supported code, chart specifications and editing. All plotted values come from the supplied AIHW workbooks.

WOMEN'S HEALTH · AUSTRALIA

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